GAMING CASTLE

Software Test Coverage Report

Backend Microservices & Angular Frontend

Test Evidence Summary

1,374 tests executed • 1,374 successful • 0 failures

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1. Executive Summary

This report consolidates the automated test execution and code-coverage evidence for the Gaming Castle application. It covers seven Spring Boot backend services and the Angular frontend. The backend results were obtained from JaCoCo reports and Maven test reports; the frontend results were obtained from the Angular test runner and its generated coverage report.

> **Overall test execution:** 1,374 tests were executed across the backend and frontend, with 1,374 successful executions and no reported failures in the supplied results.

For the backend, all seven services achieved at least 80% line coverage. The Angular frontend achieved 84.83% line coverage. Branch coverage is lower than line coverage in several components, which identifies an opportunity for additional tests around conditional paths.

2. Testing Scope

The coverage evidence included in this report covers the following implementation areas:

Layer	Components	Evidence
Backend	API Gateway; User; Booking; Payment; Tournament; Loyalty; Notification	Maven tests + JaCoCo
Frontend	Angular application, including core services, guards, interceptors, feature and shared UI	Angular/Karma coverage

3. Test Execution Summary — Backend

Service	Tests	Passed	Failures	Errors	Skipped
API Gateway	23	23	0	0	0
Booking Service	149	149	0	0	0
Loyalty Service	100	100	0	0	0
Notification Service	69	69	0	0	0
Payment Service	80	80	0	0	0
Tournament Service	97	97	0	0	0
User Service	140	140	0	0	0
TOTAL	658	658	0	0	0

All seven backend services reported 0 failures, 0 errors and 0 skipped tests in the supplied Maven/Surefire results.

4. Code Coverage — Backend

Service	Instructions	Branches	Lines	Methods	Classes
API Gateway	100%	100%	100%	100%	100%
Booking Service	99.90%	91.33%	99.77%	100%	100%
Loyalty Service	98.36%	83.22%	98.30%	98.82%	100%
Notification Service	98.82%	74.29%	100%	100%	100%
Payment Service	92.16%	78.87%	94.71%	98.11%	100%

Service	Instructions	Branches	Lines	Methods	Classes
Tournament Service	96.91%	76.36%	97.96%	100%	100%
User Service	85.29%	70.63%	88.71%	93.68%	100%

The backend coverage results show 100% class coverage for every listed service. Line coverage is above 80% for every backend service, with the User Service having the lowest backend line coverage at 88.71%.

5. Backend Coverage Assessment

Service	Line Coverage	80% Line Target	Assessment
API Gateway	100%	≥ 80%	MEETS
Booking Service	99.77%	≥ 80%	MEETS
Loyalty Service	98.30%	≥ 80%	MEETS
Notification Service	100%	≥ 80%	MEETS
Payment Service	94.71%	≥ 80%	MEETS
Tournament Service	97.96%	≥ 80%	MEETS
User Service	88.71%	≥ 80%	MEETS

Note: The 80% assessment here is specifically for line coverage. Branch coverage is reported separately and is not treated as a substitute for line coverage.

6. Test Execution Summary — Angular Frontend

Metric	Result
Tests executed	716
Successful	716
Failures	0

7. Code Coverage — Angular Frontend

Metric	Coverage	Covered / Total
Statements	82.58%	4,405 / 5,334
Branches	61.70%	1,779 / 2,883
Functions	80.65%	1,017 / 1,261
Lines	84.83%	4,057 / 4,782

The Angular test run completed with 716 of 716 tests successful. The generated coverage directory contains the main HTML dashboard at **coverage/index.html** and an LCOV data file at **coverage/lcov.info**. The report also contains detailed pages for application source files and feature areas.

8. Combined Coverage Overview

Area	Tests	Passed	Failures	Line Coverage
Backend — 7 services	658	658	0	Service-specific; all ≥ 80%
Angular Frontend	716	716	0	84.83%
Combined	1,374	1,374	0	See service/layer results

9. Coverage Observations

Strengths: All backend services meet the 80% line-coverage target; all seven backend services also report 100% class coverage. The Booking Service has particularly high line coverage (99.77%) and branch coverage (91.33%). The Angular frontend has 84.83% line coverage and 80.65% function coverage.

Improvement opportunities: Branch coverage is lower in the Angular frontend (61.70%) and in the backend Notification, Payment, Tournament and User services. This suggests additional tests could be useful for alternate conditions, error paths and other unexecuted branches.

10. Evidence and Reproducibility

Backend evidence consists of JaCoCo XML/HTML coverage outputs under each service's **target/site/jacoco/** directory and Maven/Surefire test reports under **target/surefire-reports/**. The frontend evidence consists of the Angular-generated **coverage/** directory, including **index.html** and **lcov.info**.

11. Conclusion

Based on the supplied automated test results, the Gaming Castle backend and frontend have strong automated-test coverage. The backend executed 658 tests with all 658 passing, and every backend service exceeded the 80%

line-coverage target. The Angular frontend executed 716 tests with all 716 passing and achieved 84.83% line coverage.

The results provide quantitative evidence of test execution and code coverage. Further testing can focus on lower-covered branches and, separately, the performance requirements defined for the system.

Report status: Automated test and coverage evidence consolidated successfully.